

Notes: Klapper, Laeven, and Rajan (RFS, 2012)

Trade Credit Contracts

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1 Big picture

- **Question.** What determines the terms of trade credit contracts between buyers and suppliers? In particular, who receives longer payment maturities, who receives discounts for early payment, and what do these patterns imply about why trade credit exists?
- **Core idea.** Trade credit is not only a financing instrument. It can also be a tool for buyer market power, price discrimination, product-quality warranties, and supplier risk management.
- **Why contract-level data matter.** Most prior studies use one-sided firm data, such as accounts payable or survey responses. This paper observes actual buyer-supplier contracts and therefore can examine both sides of the transaction.
- **Empirical setting.** The paper uses a unique PrimeRevenue data set containing 29,019 trade credit contracts between 56 large buyers and 24,140 suppliers, with terms such as contract amount, net days, discount days, and discount rate.
- **Headline result.** The largest and most creditworthy buyers receive the longest maturities, especially from smaller suppliers. Early-payment discounts are more likely to be offered to smaller and riskier buyers.
- **Main interpretation.** The patterns are difficult to reconcile with a pure financing story in which strong suppliers finance weak buyers. Instead, the evidence points to multiple motives: large buyers may have bargaining power, small suppliers may need to warranty product quality, and discounts help suppliers manage credit risk from riskier buyers.

2 Incremental contribution of the paper

- **Bilateral contract data.** The paper observes both buyer and supplier characteristics for the same contract. This allows the authors to study buyer-supplier pairs rather than only buyer-level or supplier-level averages.
- **Multiple contracts per party.** The data contain multiple contracts for many buyers and suppliers. This makes it possible to use buyer fixed effects and supplier fixed effects, absorbing unobserved time-invariant buyer or supplier characteristics.
- **Actual contract terms.** Instead of accounts payable or survey-based offered terms, the paper observes realized trade credit terms: net days, discount availability, discount days, discount rate, and implied effective interest rate.
- **New evidence on nonfinancial motives.** Because many buyers are large and creditworthy, the paper is not a setting where financing-constrained buyers dominate. This helps highlight product-quality warranties, buyer bargaining power, and supplier risk management.
- **Separation of maturity and discount terms.** The paper shows that long maturities and early-payment discounts appear to serve different purposes. Long maturities are often given to powerful or creditworthy buyers; discounts are targeted more to riskier buyers.
- **Challenge to a simple financing theory.** Small suppliers extending long terms to large, investment-grade buyers is not what a basic financing story would predict. This is the paper's key empirical puzzle and contribution.

3 Theories of trade credit

3.1 Financing motive

The traditional financing view is that suppliers may finance buyers when banks cannot or will not. Suppliers may have better information about buyers, greater ability to monitor them, or greater ability to liquidate goods in default. Under this motive, large and creditworthy suppliers should be especially likely to provide longer trade credit to smaller or riskier buyers.

Interpretation: In the paper’s sample, this motive has limited explanatory power because many buyers are very large and often more creditworthy than suppliers. The fact that small suppliers extend long terms to large buyers works against a simple supplier-financing story.

3.2 Buyer market power and price discrimination

Trade credit can be used as a way for large buyers to extract favorable terms from suppliers. If upfront price discrimination is difficult or legally constrained, varying the payment period may be a substitute way to favor powerful buyers.

Economic message: A market-power story predicts that large buyers should receive better terms, especially from smaller suppliers. The paper finds strong evidence consistent with this, but it also asks why powerful buyers would demand financing from suppliers that are less able to provide it, rather than simply demand a price discount.

3.3 Product-quality warranty

Trade credit may let buyers verify product quality before payment. A supplier who is less known or less trusted may extend longer payment periods to assure the buyer that the goods will perform as promised.

Interpretation: This motive is particularly relevant for small suppliers and for buyers that need time to distribute or inspect goods. It can explain why small suppliers grant longer net days to large buyers even when financing is inefficient.

3.4 Risk management and early-payment discounts

If suppliers extend credit for nonfinancial reasons, they still face default risk. Early-payment discounts can reduce that risk by encouraging risky buyers to pay quickly. This motive predicts discounts for lower-credit-quality buyers, not necessarily for the largest and safest buyers.

Takeaway: The paper’s final interpretation is a mix: long net days can reflect market power or warranty concerns, while discounts act as a risk-management tool.

4 Data and measurement

4.1 PrimeRevenue contract data

The data come from PrimeRevenue, an online supply-chain finance platform. The data are a snapshot of outstanding receivables as of December 1, 2005, before PrimeRevenue started factoring the receivables. The sample contains 29,019 contracts between 56 large buyers and 24,140 suppliers.

Key contract variables include:

- **Contract size:** amount of trade credit in U.S. dollars.
- **Net days:** number of days before payment is due.
- **Discount dummy:** indicator for whether the contract offers an early-payment discount.
- **Discount days:** number of days within which the buyer must pay to receive the discount.
- **Discount rate:** percent discount offered for early payment.
- **Effective interest rate:** implied annualized interest rate from forgoing the discount and paying later.

The effective interest rate is computed as

$$\left(\frac{1}{1 - \text{discount rate}} \right)^{\frac{360}{\text{net days} - \text{discount days}}} - 1,$$

with special treatment for contracts where net days equal discount days.

4.2 Buyer and supplier characteristics

The buyer data include size buckets, location, industry, and investment-grade status. The supplier data include size buckets and investment-grade status. Buyers are generally large and creditworthy: 59% of buyers have more than \$10 billion in sales, and 75% are investment grade. Suppliers are much smaller: 56% have less than \$100 million in sales, and 68% are not investment grade or are unrated.

5 Empirical strategy

5.1 Net days regressions

The first main outcome is the maturity of trade credit, measured by log net days. A stylized specification is

$$\ln(\text{NetDays}_{bs}) = \beta_1 \text{Buyer}_b + \beta_2 \text{Supplier}_s + \beta_3 \text{Industry}_b + \varepsilon_{bs},$$

where b denotes the buyer and s denotes the supplier. The authors estimate versions with buyer-clustered standard errors, buyer fixed effects, and supplier fixed effects.

Interpretation: The supplier fixed-effect specifications are especially informative because they compare contracts offered by the same supplier to different buyers. This asks: for a given supplier, which buyers receive longer terms?

5.2 Buyer-supplier pair interactions

To distinguish financing, bargaining, warranty, and risk-management mechanisms, the paper estimates interaction models such as supplier size interacted with buyer investment-grade status, supplier size interacted with buyer size, and supplier investment-grade status interacted with buyer characteristics.

Economic message: These interactions test whether small suppliers especially extend longer credit to powerful buyers, or whether creditworthy suppliers especially extend longer credit to risky buyers.

5.3 Discount regressions

For the discount dummy, the authors estimate logit models:

$$\Pr(\textit{Discount}_{bs} = 1) = \Lambda(X_b, X_s, \textit{Industry}_b, \text{fixed effects}),$$

where $\Lambda(\cdot)$ is the logistic function.

Interpretation: Discounts are a separate contract term from net days. If discounts are equivalent to price concessions, powerful buyers should receive them. If discounts manage default risk, riskier buyers should receive them.

5.4 Interest-rate regressions

For contracts with discounts, the paper studies the natural log of the implied effective interest rate. These regressions reveal whether discount terms are mainly determined by buyer risk, supplier characteristics, or industry norms.

6 Main findings

- **Contract maturities are long.** The mean net days is 59.2 and the median is 60, longer than the typical 30-day contract often discussed in prior work.
- **Large and investment-grade buyers get longer net days.** This is consistent with buyer market power or product-warranty motives, but not with a simple financing story.
- **Smaller suppliers extend longer terms.** This is consistent with small suppliers needing to assure product quality or being squeezed by powerful buyers.
- **Discounts are not concentrated among the most powerful buyers.** Early-payment discounts are more common for smaller and non-investment-grade buyers, suggesting that discounts help suppliers reduce default risk.
- **Industry norms are important.** Net days, discounts, and effective rates vary strongly by buyer industry. Perishable-goods sectors such as groceries and soft goods have distinctive discount and effective-interest-rate patterns.
- **No single theory fits everything.** Financing, bargaining, warranty, and risk-management motives each explain parts of the data, but none alone explains all observed contract terms.

7 Original figures and tables with reading notes

7.1 Table 1: Buyer and seller characteristics - Panels A and B and Table 1: Buyer and seller characteristics - Panel C

Table 1
Buyer and seller characteristics

Panel A: Buyer Characteristics (Percentages)						
	Number of Buyers	Number of Contracts	% of Contracts	Total Amount (\$ billion)	% of Total Amount	
Size >\$10 billion	33	39	24,298	84	612	89
Size \$0.1-10 billion	23	41	4,721	16	79	11
Industry auto	9	16	1,615	6	76	11
Industry diversified retail	13	13	9,749	34	193	28
Industry diversified manufacturing	16	29	3,824	13	74	11
Industry grocery	4	4	1,630	6	89	13
Industry hard goods retail	9	16	3,146	11	164	24
Industry soft goods retail	6	11	2,362	8	42	6
Industry technology	3	5	5,306	18	24	4
Industry food & beverages	2	4	682	2	27	4
Industry utility	1	2	705	2	2	0
Location: Europe	13	23	12,029	41	241	35
Location: North America	43	77	16,990	59	450	65
Investment Grade: No	14	25	4,008	14	107	16
Investment Grade: Yes	42	75	25,011	86	583	84
Panel B: Supplier Characteristics (Percentages)						
	Number of Suppliers	Number of Contracts	% of Contracts	Total Amount (\$ billion)	% of Total Amount	
Size >\$2 billion	2,727	11	5,772	20	531	77
Size \$0.1-2 billion	7,821	32	9,549	33	142	21
Size <\$0.1 billion	13,900	56	13,698	47	18	3
Investment Grade: No	16,391	68	18,655	65	319	46
Investment Grade: Yes	7,713	32	10,043	35	372	54

(continued)

Table 1
Continued

	Panel C: Contract Characteristics					
	N	Mean	Median	Min	Max	Std. Dev.
Contract amount (US\$ million)	29,019	23.8	3.47	0.0004	6,520.0	111.0
Net Days	29,019	59.2	60	1	120	26.1
Discount Offered (Yes/No)	29,019	0.13	0	0	1	0.33
Discount Days	3,462	30	30	1	180	20
Discount Rate (%)	3,707	2	2	.02	11.5	0.09
Ratio of Discount to Net Days	2,634	0.63	0.6	0.02	1	0.28
Effective Interest Rate (%)	2,624	54	31	2	100	38

This table reports summary statistics of buyer, supplier, and contract characteristics. Sample consists of 29,019 trade credit contracts between 56 large buyers and 24,140 suppliers. Early payment discounts are offered on 3,717 of these contracts. The effective interest rate variable is winsorized at 100% and computed after setting the spread between net days and discount days to one for contracts with net days equal to discount days. Panel A reports summary statistics for buyer characteristics, Panel B for supplier characteristics, and Panel C for contract characteristics.

Table 1: Buyer and seller characteristics - Panel C

Table 1: Buyer and seller characteristics - Panels A and B

Read: Table 1 describes the sample. Panels A and B show that buyers are large and creditworthy, while suppliers are smaller and often not investment grade. Panel C summarizes the contract terms: the mean net days is 59.2, 13% of contracts have a discount, and the median discount rate is 2%.

Economic message: The table sets up the core puzzle. If suppliers mainly finance constrained buyers, it is surprising that many small suppliers extend credit to very large buyers.

7.2 Table 2: Buyer and seller cross-tabulations and Table 3: Distribution of buyer and seller characteristics, by contract characteristics

Table 2
Buyer and seller cross-tabulations

Panel A: Buyer vs. Supplier Characteristics (number of observations)						
	Supplier Size			Supplier Investment Grade		Total
	Small	Medium	Large	No	Yes	
Buyer Size						
Small/Medium	1,608	2,166	947	3,471	1,250	4,721
Large	12,090	7,383	4,825	15,505	8,793	24,298
Buyer Investment Grade						
No	1,204	1,729	1,075	2,667	1,341	4,008
Yes	12,494	7,820	4,697	16,309	8,702	25,011
Total	13,698	9,549	5,772	18,976	10,043	29,019
Panel B: Buyer Characteristics (number of buyers)						
	Buyer size			Total		
	Small /Medium	Large				
Buyer Investment Grade						
No		6	8	14		
Yes		17	25	42		
Total		23	33	56		

This table reports cross-tabulations of count statistics of buyer and supplier characteristics (Panel A) and within-buyer characteristics (Panel B). Sample consists of 29,019 trade credit contracts between 56 large buyers and 24,140 suppliers. Not investment grade also includes unrated firms.

Table 2: Buyer and seller cross-tabulations

Read: Table 2 shows the distribution of buyer-supplier matches by size and investment grade. Table 3 links contract amount, net days, and location to buyer and supplier characteristics.

Table 3
Distribution of buyer and seller characteristics, by contract characteristics (percentages)

	Contract Amount (%)				Net Days (%)				Location	
	< \$1 million	\$1-4 million	\$4-15 million	> \$15 million	0-30	31-60	61-90	91+	Europe	NA
<i>Buyer Characteristics:</i>										
Size >\$10 billion	32	23	20	25	20	38	29	13	46	54
Size \$0.1-10 billion	5	35	33	26	52	40	8	0	18	82
Industry auto	0	15	28	57	20	53	20	7	59	41
Industry diversified retail	23	37	22	17	10	13	50	26	84	16
Industry diversified mfg.	6	39	36	29	47	34	17	1	0	100
Industry grocery	11	25	18	47	42	54	3	0	84	16
Industry hard goods retail	0	15	29	57	21	52	24	4	3	97
Industry soft goods retail	26	23	22	29	85	14	1	1	0	100
Industry technology	74	9	10	7	8	88	3	0	0	100
Industry food & beverage	28	34	21	17	27	30	12	31	100	0
Industry utility	73	18	6	4	54	9	36	0	100	0
Location: Europe	26	33	20	20	11	19	46	23	100	0
Location: North America	29	19	23	29	37	53	10	1	0	100
Investment Grade: No	20	20	28	32	68	29	2	0	7	93
Investment Grade: Yes	29	26	21	24	18	39	26	12	50	50
<i>Supplier Characteristics:</i>										
Size >\$2 billion	6	11	14	64	13	30	74	9	30	70

Table 3: Distribution of buyer and seller characteristics, by contract characteristics

Interpretation: Table 3 shows that contract terms differ sharply across buyer and supplier types. Large buyers account for most contracts and dollar volume, while large suppliers account for most dollar volume despite being a smaller fraction of suppliers.

7.3 Table 4: Distribution of buyer and seller characteristics, by discount characteristics and Table 5: Summary statistics of regression variables

Table 4
Distribution of buyer and seller characteristics, by discount characteristics (percentages)

	Full Sample	Subsample of Contracts that Offer an Early Payment Discount						
		Discount Rate (%)			Discount Days (%)		Discount to Net Days Ratio (%)	
	Discount (%)	0–1%	1–2%	> 2%	0–30	31–60		61+
<i>Buyer Characteristics:</i>								
Size >\$10 billion	10	35	58	7	64	33	3	64
Size \$0.1–10 billion	26	37	52	11	78	21	2	60
Industry auto	19	21	50	29	100	0	0	35
Industry diversified retail	5	34	66	0	94	5	1	43
Industry diversified mfg	13	67	30	3	95	4	1	44
Industry grocery	25	31	48	22	84	15	0	87
Industry hard goods retail	58	35	60	5	54	42	4	68
Industry soft goods retail	8	5	86	9	25	75	0	95
Location: Europe	4	19	44	37	70	28	2	80
Location: North America	19	38	58	4	67	30	3	61
Investment Grade: No	17	42	57	1	93	6	1	53
Investment Grade: Yes	12	34	56	10	57	32	2	66
<i>Supplier Characteristics:</i>								
Size >\$2 billion	27	34	58	7	66	31	3	65
Size \$0.1–2 billion	17	38	54	8	67	30	2	63
Size <\$0.1 billion	4	32	58	11	80	19	1	57
Investment Grade: No	13	37	55	9	69	29	2	63
Investment Grade: Yes	12	33	59	7	67	30	3	64

This table reports the distribution of trade credit contract terms (in percentages) by buyer and supplier

Table 4: Distribution of buyer and seller characteristics, by discount characteristics

Table 5
Summary statistics of regression variables

Variable	Obs	Mean	Std. Dev.	Min	Max
<i>Complete sample:</i>					
Log net days	28,187	3.947	0.614	0	5.481
Discount dummy	29,019	0.128	0.334	0	1
<i>Subsample of contracts with early payment discount:</i>					
Discount days	3,462	30.433	20.090	1	180
Discount rate	3,707	0.017	0.008	0.000	0.115
Discount days/Net days	2,634	0.630	0.282	0.017	1
Effective rate	2,584	0.533	0.381	0.017	1
<i>Buyer characteristics:</i>					
Buyer large size	56	0.589	0.496	0	1
Buyer small size	56	0.411	0.496	0	1
Buyer investment grade	56	0.750	0.437	0	1
Buyer North America	56	0.768	0.426	0	1
Industry auto	56	0.161	0.371	0	1
Industry diversified retail	56	0.125	0.334	0	1
Industry diversified mfg	56	0.286	0.456	0	1
Industry grocery	56	0.071	0.260	0	1
Industry hard goods retail	56	0.150	0.354	0	1
Industry soft goods retail	56	0.100	0.296	0	1
Industry technology	56	0.054	0.227	0	1
Industry food and beverages	56	0.036	0.187	0	1
Industry utility	56	0.018	0.134	0	1
<i>Supplier characteristics:</i>					
Supplier large size	24,140	0.113	0.316	0	1
Supplier medium size	24,140	0.324	0.468	0	1
Supplier small size	24,140	0.563	0.496	0	1
Supplier investment grade	24,140	0.320	0.466	0	1

This table reports summary statistics of the main regression variables. Sample consists of 29,019 trade credit contracts between 56 large buyers and 24,140 suppliers. Early payment discounts are offered on 3,717 of these contracts. Summary statistics on buyer-specific variables are computed at the buyer level, and summary statistics on supplier-specific variables are computed at the supplier level.

Table 5: Summary statistics of regression variables

Read: Table 4 summarizes the subsample with early-payment discounts. Table 5 provides the regression-variable summary statistics.

Takeaway: Discounts are not the dominant contract form: they appear in about 13% of contracts. But when they appear, the implied effective interest rates can be very high, making discounts economically important.

7.4 Table 6: Correlation matrix of regression variables - Panel A and Table 6: Correlation matrix of regression variables - Panel B

Table 6
Correlation matrix of regression variables

Panel A: Correlations between credit terms, buyer characteristics, and supplier characteristics

	Full Sample		Subsample w/Discount	
	Log net days	Discount dummy	Discount days/net days	Effective rate
Buyer large size	0.26*	-0.18*	0.09*	0.01
Buyer small size	-0.26*	0.18*	-0.09*	-0.02
Buyer investment grade	0.29*	-0.05*	0.20*	0.15*
Buyer North America	-0.37*	0.22*	-0.18*	-0.26*
Industry auto	0.00	0.04*	-0.17*	-0.12*
Industry diversified retail	0.35*	-0.17*	-0.33*	-0.27*
Industry diversified mfg	-0.13*	0.00	-0.32*	-0.26*
Industry grocery	-0.17*	0.09*	0.35*	0.44*
Industry hard goods retail	-0.01	0.47*	0.15*	-0.03
Industry soft goods retail	-0.25*	-0.04*	0.33*	0.34*
Industry technology	-0.02*	-0.18*	n.a.	n.a.
Industry food and beverage	0.05*	-0.06*	n.a.	n.a.
Industry utility	-0.07*	-0.06*	n.a.	n.a.
Supplier large size	-0.11*	0.21*	0.06*	0.08*
Supplier medium size	-0.10*	0.08*	0.00	-0.05
Supplier small size	0.18*	-0.25*	-0.08*	-0.04
Supplier investment grade	0.03*	-0.02*	0.02	0.02

(continued)

Table 6
Continued

Panel B: Correlations between buyer and supplier characteristics

	Buyer large size	Buyer investment grade	Buyer North America	Industry auto	Industry diversified retail	Industry diversified mfg	Industry grocery	Industry hard goods retail	Industry soft goods retail	Industry technology	Industry food and beverage	Industry utility	Supplier large size	Supplier medium size
Buyer investment grade	0.02													
Buyer North America	-0.12	-0.12												
Industry auto	-0.03	-0.20	-0.10											
Industry diversified retail	0.10	0.09	-0.18	-0.17										
Industry diversified mfg	-0.28	0.00	0.35*	-0.28	-0.24									
Industry grocery	0.23	-0.00	-0.34	-0.12	-0.10	-0.18								
Industry hard goods retail	-0.06	0.13	0.11	-0.19	-0.16	-0.27	-0.12							
Industry soft goods retail	0.04	0.06	0.19	-0.15	-0.13	-0.22	-0.09	-0.10						
Industry technology	0.20	-0.05	0.13	-0.10	-0.09	-0.15	-0.07	-0.10	-0.08					
Industry food and beverage	-0.03	-0.11	-0.35	-0.08	-0.07	-0.12	-0.05	-0.08	-0.07	-0.05				
Industry utility	0.11	0.08	-0.25	-0.06	-0.05	-0.09	-0.04	-0.06	-0.05	-0.03	-0.03			
Supplier large size	-0.02	-0.07*	0.09*	0.11*	-0.10*	0.02*	0.11*	0.21*	-0.02*	-0.14*	-0.00	-0.08*		
Supplier medium size	-0.14*	-0.10*	0.06*	0.11*	-0.06*	0.14*	0.02	0.12*	0.05*	-0.21*	-0.02	-0.08*	-0.25*	
Supplier investment grade	0.08*	0.01	-0.04*	0.02*	0.03*	-0.04*	-0.02*	0.00	-0.05*	0.04*	-0.03*	0.01	0.11*	-0.05*

This table reports correlations between the main regression variables. Panel A presents correlations between the explanatory and dependent variables. Panel B presents correlations between the explanatory variables. Sample consists of 29,039 trade credit contracts between 96 large buyers and 24,140 suppliers. Early payment discounts are offered on 3,717 of these contracts. Correlations between buyer characteristics are computed at the buyer level, correlations between supplier characteristics are computed at the supplier level, and correlations between

Table 6: Correlation matrix of regression variables - Panel B

Table 6: Correlation matrix of regression variables - Panel A

Read: Table 6 reports correlations among contract terms, buyer characteristics, and supplier characteristics.

Interpretation: The low correlation between size and investment-grade status helps the regressions separately identify size effects and creditworthiness effects. The correlations also preview that industry and location are important determinants of contract terms.

7.5 Table 7: Variance decomposition analysis - Panel A and Table 7: Variance decomposition analysis - Panel B

Table 7
Variance decomposition analysis

Panel A: Full sample					
Dependent variable: Log of net days	Degrees of freedom	Partial sum of squares	F-stat	p-value	Contribution to sum of squares (%)
Buyer's size	1	127.94	509.71	0.000	3.60
Buyer's sub-industry	8	1145.83	570.64	0.000	32.28
Buyer's location	3	458.36	608.72	0.000	12.91
Buyer's investment grade	1	117.80	469.34	0.000	3.32
Supplier's size	2	61.83	123.17	0.000	1.74
Supplier's investment grade	1	1.36	5.42	0.020	0.04
Number of obs. = 28,187 Adj. R-squared = 0.334					
Dependent variable: Discount dummy	Degrees of freedom	Partial sum of squares	F-stat	p-value	Contribution to sum of squares (%)
Buyer's size	1	9.08	123.38	0.264	0.82
Buyer's sub-industry	8	738.17	1253.48	0.000	66.74
Buyer's location	3	178.13	806.64	0.000	16.11
Buyer's investment grade	1	0.00	0.00	0.980	0.00
Supplier's size	2	0.45	3.04	0.048	0.04
Supplier's investment grade	1	0.21	2.84	0.092	0.02
Number of obs. = 29,019 Adj. R-squared = 0.341					
Dependent variable: Effective interest rate	Degrees of freedom	Partial sum of squares	F-stat	p-value	Contribution to sum of squares (%)
Buyer's size	1	3.18	37.97	0.000	1.93
Buyer's sub-industry	5	131.15	312.76	0.000	79.42
Buyer's location	2	9.27	55.25	0.000	5.61
Buyer's investment grade	1	5.36	63.93	0.000	3.25
Supplier's size	2	1.80	10.75	0.000	1.09
Supplier's investment grade	1	0.00	0.05	0.820	0.00
Number of obs. = 2,624 Adj. R-squared = 0.427					
Panel B: Supplier fixed effects					
Dependent variable: Log of net days	Degrees of freedom	Partial sum of squares	F-stat	p-value	Contribution to sum of squares (%)
Buyer's size	1	16.25	62.14	0.000	0.67
Buyer's sub-industry	8	73.93	35.35	0.000	3.07
Buyer's location	3	35.03	44.66	0.000	1.45
Buyer's investment grade	1	18.24	69.77	0.000	0.76
Supplier fixed effect	2266	1371.56	2.32	0.000	56.89
Number of obs. = 6,448 Adj. R-squared = 0.519					

(continued)

Table 7
Continued

Panel B: Supplier fixed effects					
Dependent variable: Discount dummy	Degrees of freedom	Partial sum of squares	F-stat	p-value	Contribution to sum of squares (%)
Buyer's size	1	4.76	67.82	0.450	0.44
Buyer's sub-industry	8	46.82	83.31	0.000	4.31
Buyer's location	3	18.39	87.26	0.000	1.69
Buyer's investment grade	1	0.56	8.05	0.005	0.05
Supplier fixed effect	2393	483.93	2.88	0.000	44.57
Number of obs. = 7,273 Adj. R-squared = 0.642					
Dependent variable: Effective interest rate	Degrees of freedom	Partial sum of squares	F-stat	p-value	Contribution to sum of squares (%)
Buyer's size	1	0.01	0.09	0.769	0.00
Buyer's sub-industry	5	9.96	33.14	0.000	8.40
Buyer's location	2	2.66	22.13	0.000	2.24
Buyer's investment grade	1	2.57	42.73	0.000	2.17
Supplier fixed effect	539	52.44	1.62	0.000	44.22
Number of obs. = 1,049 Adj. R-squared = 0.576					

This table shows the results of an ANOVA analysis of variance of the log of net days, discount dummy, and effective interest rate variables. Specifically, we report the contribution of buyer and supplier characteristics in

Table 7: Variance decomposition analysis - Panel B

Table 7: Variance decomposition analysis - Panel A

Read: Table 7 decomposes variation in log net days, the discount dummy, and effective interest rates across buyer and supplier characteristics.

Economic message: Buyer industry explains a large fraction of variation in contract terms, especially discounts and interest rates. Once supplier fixed effects are included, buyer characteristics still matter for discounts and effective rates, but less for net days.

7.6 Figure 1: Net days for suppliers and buyers of different size and ratings and Table 8: Net days

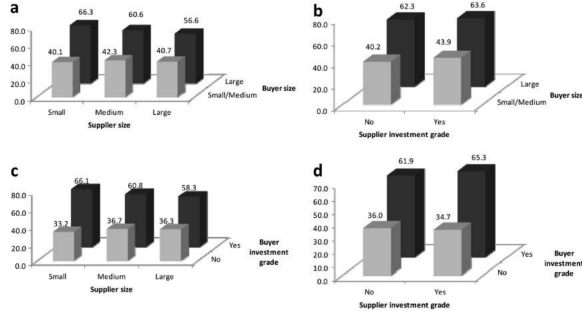


Figure 1
Net days for suppliers and buyers of different size and ratings
Figures 1a through 1d report the average net days for different subgroups of supplier-buyer pairs based on supplier size, supplier creditworthiness, buyer size, and buyer creditworthiness.

Figure 1: Net days for suppliers and buyers of different size and ratings

Table 8
Net days

Dependent variable: Log net days	Buyer clustered		Buyer FE		Supplier FE	
	Without discount		Without discount		Without discount	
	(1)	(2)	(3)	(4)	(5)	(6)
Buyer large size	0.387*** (0.102)	0.388*** (0.110)			0.225*** (0.025)	0.172*** (0.031)
Buyer investment grade	0.238** (0.105)	0.293*** (0.113)			0.226*** (0.025)	0.280*** (0.030)
Buyer North America	-0.488*** (0.168)	-0.556*** (0.184)			-0.443*** (0.043)	-0.499*** (0.054)
Industry diversified retail	-0.071 (0.127)	-0.136 (0.155)			0.036 (0.060)	-0.067 (0.071)
Industry diversified mfg	0.095 (0.166)	0.124 (0.186)			0.181*** (0.052)	0.169*** (0.060)
Industry grocery	-0.763*** (0.136)	-0.760*** (0.163)			-0.534*** (0.069)	-0.535*** (0.084)
Industry hard goods retail	0.134 (0.150)	0.110 (0.166)			0.067 (0.061)	-0.001 (0.073)
Industry soft goods retail	-0.262* (0.143)	-0.230 (0.161)			-0.125 (0.086)	-0.103 (0.097)
Industry technology	-0.048 (0.304)	-0.044 (0.321)			-0.237*** (0.062)	-0.287*** (0.069)
Industry food and beverage	-0.148 (0.143)	-0.200 (0.158)			0.100 (0.131)	0.043 (0.141)
Industry utility	-0.805*** (0.122)	-0.872*** (0.156)			-0.552*** (0.119)	-0.661*** (0.128)
Supplier small size	0.149 (0.126)	0.145 (0.138)	0.059*** (0.008)	0.068*** (0.008)		
Supplier medium size	0.009 (0.009)	-0.011 (0.009)	0.018** (0.008)	0.022*** (0.008)		

Table 8: Net days

Read: Figure 1 plots average net days across buyer-supplier groups. Table 8 estimates regressions for log net days with buyer-clustered errors, buyer fixed effects, and supplier fixed effects.

Interpretation: The central pattern is that larger and investment-grade buyers get longer terms. In supplier fixed-effect regressions, large buyers and investment-grade buyers continue to receive longer net days from the same supplier. Small suppliers also extend longer terms in buyer fixed-effect specifications.

7.7 Table 9: Heterogeneous effects of buyer and supplier characteristics and Figure 2: Discount frequency for suppliers and buyers of different size and ratings

Table 9

Heterogeneous effects of buyer and supplier characteristics

Dependent variable: Log net days	(1)	(2)	(3)	(4)	(5)
Supplier large size	-0.128*** (0.020)	0.007 (0.018)	-0.058*** (0.008)	-0.061*** (0.008)	-0.056** (0.026)
Supplier medium size	-0.060*** (0.017)	-0.048*** (0.015)	-0.042*** (0.006)	-0.042*** (0.006)	-0.070*** (0.022)
Supplier invst grade	0.017*** (0.005)	0.016*** (0.005)	-0.028** (0.013)	0.047*** (0.014)	-0.006 (0.019)
Supplier large size * Buyer invst grade	0.082*** (0.022)				0.072*** (0.022)
Supplier medium size * Buyer invst grade	0.020 (0.019)				0.020 (0.019)
Supplier large size * Buyer large size		-0.084*** (0.019)			-0.079*** (0.021)
Supplier medium size * Buyer large size		0.009 (0.016)			0.012 (0.017)
Supplier invst grade * Buyer invst grade			0.053*** (0.014)		0.038** (0.015)
Supplier invst grade * Buyer large size				-0.035** (0.014)	-0.012 (0.015)
No. of buyers	56	56	56	56	56
No. of observations	28,187	28,187	28,187	28,187	28,187
R-squared	0.003	0.004	0.003	0.003	0.004

In columns (1) through (3), the dependent variable is log net days and regression estimates are based on an OLS model with buyer fixed effects. In columns (4) through (6), the dependent variable is a dummy variable that take a value of one if the trade credit contract includes a discount (two-part contract), and zero otherwise, and regression estimates are based on a logit model with buyer fixed effects. All columns include buyer fixed effects.

Table 9: Heterogeneous effects of buyer and supplier characteristics

Read: Table 9 studies interactions between buyer and supplier size and creditworthiness. Figure 2 plots the frequency of early-payment discounts across buyer-supplier groups.

Economic message: Table 9 shows that small suppliers tend to offer longer terms. Figure 2 shows that discounts are more common for small and non-investment-grade buyers, especially from smaller or weaker suppliers.

Takeaway: This combination supports a two-part interpretation: long maturity can be a warranty or bargaining outcome, while discounts are used to reduce the risk of lending to weaker buyers.

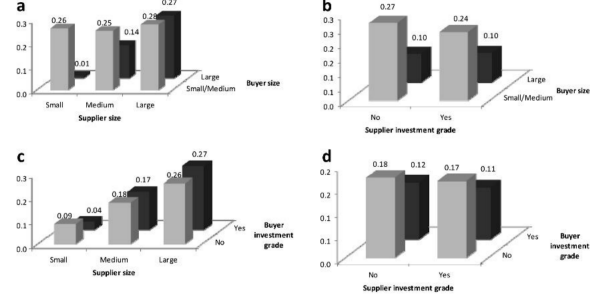


Figure 2

Discount frequency for suppliers and buyers of different size and ratings

Figures 2a through 2d report the fraction of contracts that receive discounts for different subgroups of supplier-buyer pairs based on supplier size, supplier creditworthiness, buyer size, and buyer creditworthiness.

Figure 2: Discount frequency for suppliers and buyers of different size and ratings

7.8 Table 10: Discounts and Table 11: Interest rates

Table 10
Discounts

Dependent variable: Discount	Buyer clustered		Buyer FE	Supplier FE	
	Buyers who have at least one discount			Buyers who have at least one discount	
	(1)	(2)	(3)	(4)	(5)
Buyer large size	-1.587** (0.698)	-0.738 (0.504)		-1.673*** (0.179)	-1.313*** (0.190)
Buyer investment grade	-0.546 (0.681)	-0.568 (0.684)		-0.779*** (0.174)	-0.835*** (0.214)
Buyer North America	1.753 (1.270)	0.839 (1.079)		3.365*** (0.386)	3.667*** (0.527)
Industry diversified retail	0.929 (1.428)	-0.790 (1.042)		2.504*** (0.419)	0.647 (0.516)
Industry diversified mfg	-0.005 (1.097)	-1.706* (0.947)		-0.148 (0.336)	-1.442*** (0.464)
Industry grocery	3.189* (1.657)	-0.152 (1.129)		5.632*** (0.559)	3.373*** (0.710)
Industry hard goods retail	2.937*** (0.963)	0.472 (1.061)		3.838*** (0.387)	1.644*** (0.484)
Industry soft goods retail	-0.068 (1.130)	-1.364 (1.033)		0.647 (0.753)	-0.466 (0.862)
Supplier small size	-1.215*** (0.344)	-0.601 (0.419)	0.225** (0.095)		
Supplier medium size	-0.466*** (0.158)	-0.409*** (0.175)	-0.098 (0.066)		
Supplier investment grade	-0.280*** (0.093)	-0.099 (0.107)	-0.065 (0.064)		
Number of buyers	56	34	34	56	34
Number of suppliers	24,140	7,927	7,927	399	305
Number of observations	29,019	10,604	10,604	2,067	1,433
Pseudo R-squared	0.336	0.118	0.002	0.295	0.150

Dependent variable is a dummy variable that takes a value of one if the trade credit contract includes a discount (two-part contract), and zero otherwise. Regression estimates are based on a logit model. Standard errors in regressions (1) through (2) are corrected for clustering at the buyer level. Regression (3) includes buyer fixed effects (note that by definition, this sample is equivalent to the sample of buyers who have at least one discount). Regressions (4) and (5) include supplier fixed effects and are estimated based on the subsample of suppliers that have multiple contracts. Regressions (2) and (5) include only buyers who have at least one discount. Several industries do not have firms with discounts and are dropped from estimation. Standard errors are reported between brackets. ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively.

That early payment discounts are more common from small suppliers is consistent with the market power hypothesis, except that these discounts go more to small and non-investment-grade buyers. The finding is more consistent with the risk management view that stipulates that smaller suppliers are more likely to offer discounts to risky buyers as a way to encourage early payment and prevent default because it is more difficult for these firms to absorb and diversify default risk.¹⁵

Table 10: Discounts

Table 11
Interest rates

Dependent variable:	Effective interest rate			Effective interest rate with spread set to one if net days equals discount days		
	Buyer clustered (1)	Buyer FE (2)	Supplier FE (3)	Buyer clustered (1)	Buyer FE (2)	Supplier FE (3)
Buyer large size	-0.177 (0.473)		0.043 (0.093)	-0.174 (0.461)		0.076 (0.092)
Buyer investment grade	0.396* (0.205)		0.666*** (0.105)	0.397* (0.203)		0.676*** (0.105)
Buyer North America	0.622*** (0.202)		0.806*** (0.240)	0.622*** (0.201)		0.811*** (0.241)
Industry diversified retail	0.190 (0.311)	-0.241 (0.309)	0.190 (0.309)	0.190 (0.309)	-0.230 (0.310)	0.190 (0.310)
Industry diversified mfg	-0.211 (0.375)	-0.177 (0.277)	-0.210 (0.372)	-0.210 (0.372)	-0.157 (0.279)	-0.157 (0.279)
Industry grocery	1.643*** (0.303)	0.972*** (0.320)	1.643*** (0.301)	0.972*** (0.301)	0.982*** (0.321)	0.982*** (0.321)
Industry hard goods retail	0.171 (0.464)	-0.667*** (0.305)	0.206 (0.458)	0.206 (0.458)	-0.655*** (0.305)	-0.655*** (0.305)
Industry soft goods retail	1.078*** (0.352)	0.620 (0.535)	1.083*** (0.343)	0.620 (0.535)	0.626 (0.537)	0.626 (0.537)
Supplier small size	0.099 (0.137)	-0.089** (0.036)	0.101 (0.134)	-0.084** (0.037)		
Supplier medium size	-0.069* (0.036)	-0.062*** (0.024)	-0.067* (0.037)	-0.054** (0.024)		
Supplier investment grade	0.035 (0.064)	0.036 (0.024)	0.033 (0.061)	0.037 (0.024)		
Number of buyers	34	34	28	34	34	28
Number of suppliers	2,080	2,080	531	2,115	2,115	540
Number of observations	2,584	2,584	1,035	2,624	2,624	1,049
R-squared	0.353	0.007	0.216	0.347	0.005	0.213

Dependent variable is the natural logarithm of the effective interest rate on the trade credit contract, with the effective interest rate computed as $(1/(1 - \text{discount rate})^{360/(\text{net days} - \text{discount days})} - 1)$. In columns 4–6, the effective interest rate is computed by setting the spread between net days and discount days to one for contracts with net days equal to discount days (these contracts are dropped from the regressions in columns 1–3). We winsorize interest rates at 100% before taking logs. Standard errors in regressions (1) and (4) are corrected for clustering at the buyer level. Regressions (2) and (5) include buyer fixed effects (note that by definition, this sample is equivalent to the sample of buyers excluding discounts). Regressions (3) and (6) include supplier fixed effects and are estimated based on the subsample of suppliers that have multiple contracts. Several industries do not have firms with discounts and are dropped from the regressions. The regressions also exclude contracts with

Table 11: Interest rates

Read: Table 10 estimates logit regressions for whether a contract includes a discount. Table 11 studies the effective interest rate in contracts that include discounts.

Interpretation: Table 10 shows that large and investment-grade buyers are less likely to receive early-payment discounts. This is inconsistent with a pure bargaining-power story in which powerful buyers extract price discounts. Table 11 shows that discount terms are strongly industry-specific; groceries and soft goods often have high effective rates, consistent with supplier concerns about perishable goods and repayment risk.

8 How to read the paper's contribution in one pass

- Start with the contrast between buyers and suppliers: buyers are typically much larger and more creditworthy.
- Then read Table 8 and Figure 1: longer maturity flows toward large and creditworthy buyers, especially from smaller suppliers.
- Next read Figure 2 and Table 10: discounts do not flow primarily to the powerful buyers. They

are more common for smaller and riskier buyers.

- This separation between maturity and discounts is the paper’s central insight. Trade credit terms are multidimensional, and each dimension appears to solve a different problem.
- The financing story is not absent, but it is incomplete. The data point toward a richer theory of contracts in which suppliers offer credit to certify product quality and use discounts to manage credit risk.

9 Critical assessment

9.1 Strengths

- **Unique data.** The paper uses actual contract-level trade credit data, not just balance-sheet aggregates or surveys.
- **Two-sided information.** Observing both buyer and supplier characteristics allows the authors to test theories about bargaining power and relative creditworthiness.
- **Fixed effects.** Multiple contracts per buyer and supplier allow for buyer and supplier fixed effects, which sharply improves identification relative to prior work.
- **Contract multidimensionality.** The distinction between net days and discounts is economically important and generates the paper’s main interpretation.

9.2 Limitations

- **Few buyers.** There are only 56 buyers, and they are mostly large firms. The sample is not ideal for studying small, credit-constrained buyers.
- **Limited supplier information.** Supplier sector and location are unavailable, so supplier fixed effects are important but cannot answer every mechanism question.
- **Cross-sectional snapshot.** The data are a single snapshot from 2005, so the paper cannot study contract dynamics over time or responses to macro shocks.
- **Reduced-form interpretation.** The evidence documents strong associations, but causality across bargaining power, warranty motives, and credit-risk management remains difficult to pin down fully.

10 Bottom line

Klapper, Laeven, and Rajan show that trade credit contracts are not simply loans from suppliers to buyers. Contract maturity and early-payment discounts appear to serve different functions. Long net days are disproportionately granted to large, creditworthy buyers by smaller suppliers, consistent with buyer power and product-quality warranty motives. Discounts, by contrast, are directed more toward riskier buyers, consistent with supplier risk management. The paper’s main contribution is to use bilateral, multicontract data to show that trade credit is a richer contractual arrangement than a single financing channel.